

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0227

Cooper 138 kV
November 27, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0227. The Interconnection Customer proposes a 185.1 MW wind facility interconnecting at the Cooper 138 kV substation in the Kentucky area. The request was received on April 13, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 154 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$60,821,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Berry St - Wisota 138.0 kV
Contingency: Killen - Toledo End 500.0 kV line
Loading: 134.6% (Rating: 272 MVA)
DFAX: 16.2%
Pre-project loading: 123.6%
2. Facility: Universal Studios Tap - Cooper 138.0 kV
Contingency: Lee County Solid Waste Energy Substation - Cooper 345.0 kV
Loading: 117.8% (Rating: 268 MVA)
DFAX: 32.3%
Pre-project loading: 95.4%
3. Facility: Bozemansouth - Anson N 138.0 kV
Contingency: Lee County Solid Waste Energy - Cooper 345.0 kV
Loading: 116.9% (Rating: 263 MVA)
DFAX: 20.8%
Pre-project loading: 102.2%
4. Facility: Ironman - Goldsboro 230.0 kV
Contingency: Toledo Bend - Killen 500.0 kV
Loading: 115.8% (Rating: 402 MVA)
DFAX: 8.5%
Pre-project loading: 111.9%
5. Facility: Bulan - Hazard 138.0 kV Circuit 1
Contingency: Hazard - Bulan 138.0 kV Cir. 2
Loading: 115.7% (Rating: 281 MVA)
DFAX: 5.3%
Pre-project loading: 112.2%

6. Facility: Bulan Substation - Hazard 138.0 kV
Contingency: Hazard - Bulan 138.0 kV
Loading: 115.7% (Rating: 281 MVA)
DFAX: 5.3%
Pre-project loading: 112.2%
7. Facility: Atoka - Spring Crk Rd 138.0 kV
Contingency: Scott Branch - Wansley Combined Cycle 138.0 kV
Loading: 109.1% (Rating: 286 MVA)
DFAX: 6.7%
Pre-project loading: 104.8%
8. Facility: Sidney Coal - Scott Branch 138.0 kV line
Contingency: Spring Crk Road - Haris 138.0 kV line
Loading: 105.9% (Rating: 237 MVA)
DFAX: 5.4%
Pre-project loading: 101.7%
9. Facility: Scott Branch - Wansley Combined Cycle 138.0 kV
Contingency: Spring Crk Rd - Harris Substation 138.0 kV
Loading: 104.7% (Rating: 260 MVA)
DFAX: 5.4%
Pre-project loading: 100.9%
10. Facility: Kean Mtn And Grassy Creek 138-1 kV Transformer
Contingency: Clinch Riv - Carlton Hills 345.0 kV
Loading: 101.8% (Rating: 345 MVA)
DFAX: 5.3%
Pre-project loading: 98.9%
11. Facility: Greenville Ave - Beanblossom-Sw Sta 138.0 kV
Contingency: Lee County Solid Waste Energy - Cooper Substation 345.0 kV
Loading: 101.6% (Rating: 243 MVA)
DFAX: 17.8%
Pre-project loading: 88.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wissota - Berry Street 138.0 kV	Rebuild line with higher-capacity conductor	\$1,049,000	49% (\$515,000)
Universal Studios Tap - Cooper 138.0 kV	Rebuild line with higher-capacity conductor	\$4,241,000	71% (\$3,017,000)
Anson Nrth - Bozeman S Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$23,406,000	56% (\$13,136,000)
Ironan - Goldsboro 230.0 kV	Rebuild line with higher-capacity conductor	\$66,910,000	48% (\$31,944,000)
Bulan - Hazard Substation 138.0 kV #1	Rebuild line with higher-capacity conductor	\$1,781,000	100% (\$1,781,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bulan - Hazard 138.0 kV Cir. 2	Rebuild line with higher-capacity conductor	\$2,874,000	100% (\$2,874,000)
Atoka - Spring Crk Rd 138.0 kV	Reconductor line	\$6,151,000	8% (\$501,000)
Sidney Coal - Scott Branch 138.0 kV line	Reconductor line	\$2,453,000	6% (\$160,000)
Scott Branch - Wansley Combined Cycle 138.0 kV	Reconductor line	\$7,706,000	6% (\$503,000)
Kean Mtn And Grassy Crk 138/1kV TX	Replace transformer with higher-rated unit	\$6,692,000	15% (\$1,003,000)
Greenville Ave - Beanblossom-Sw Sta 138.0 kV	Reconductor line	\$11,456,000	47% (\$5,387,000)
	Total allocated to project		\$60,821,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.